III B.Tech – II Semester

Course Code	**ENVIRONMENTAL ENGINEERING LAB**	L	T	P	C
23A01603P		0	0	3	1.5

Course Objectives:
The objectives of this course are to make the student to:

1. **Understand** the principles and methods of water and wastewater sampling and preservation.
2. **Perform** standard laboratory tests to determine water quality parameters.
3. **Analyze** wastewater characteristics and assess pollution levels.
4. **Evaluate** the effectiveness of treatment processes using chemical and biological tests.
5. **Develop** hands-on skillsin advanced laboratory techniques for environmental monitoring.

Course Outcomes:
Upon successful completion of this course, students will be able to:

2. **Apply** appropriate sampling and preservation techniques for water and wastewater.
3. **Measure** physical and chemical parameters such as turbidity, conductivity, and chlorine content.
4. **Analyze** key water and wastewater quality indicators like BOD, COD, and TKN.
5. **Assess** the efficiency of water treatment processes through laboratory tests.
6. **Perform** microbiological analysis forcoli form detection and sludge characterization.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	**3**	**1**	**1**	**-**	**-**	**2**	**2**	**-**	**-**	**-**	**1**	**1**	**1**	**1**
CO -2	**2**	**2**	**1**	**-**	**-**	**1**	**1**	**-**	**-**	**-**	**1**	**1**	**2**	**2**
CO -3	**2**	**1**	**2**	**-**	**1**	**1**	**2**	**-**	**-**	**-**	**2**	**1**	**1**	**1**
CO -4	**1**	**1**	**1**	**1**	**-**	**1**	**1**	**-**	**-**	**-**	**1**	**1**	**2**	**2**
CO -5	**2**	**1**	**2**	**-**	**-**	**1**	**2**	**1**	**-**	**-**	**1**	**1**	**1**	**1**

LIST OF EXPERIEMENTS:-

II. ANALYSIS of WATER SAMPLE

1. Sampling and preservation methods for water and wastewater (Demonstration only)
2. Measurement of Electrical conductivity and turbidity
3. Determination of fluoride in water by spectrophotometric method /ISE
4. Determination of iron in water (Demo)
5. Determination ofSulphate in water
6. Determination of Optimum Coagulant Dosage by Jar test apparatus
7. Determination of available Chlorine in Bleaching powder and residual chlorine in water

III. ANALYSIS of WASTEWATER SAMPLE

1. Estimation of suspended, volatile and fixed solids
2. Determination of Sludge Volume Index in waste water
3. Determination of Dissolved Oxygen
4. Estimation of B.O.D.

5. Estimation of C.O.D.
6. Determination of TKN and Ammonia Nitrogen in wastewater
7. Determination of total andfaecal coliform (Demonstration only)

Note: Minimum 10 out of the above experiments are to be carried out.

TEXT BOOKS:

1. Manual on Water Supply and Treatment. Ministry of Urban Development, New Delhi.
2. Manual on Sewerage and Sewage Treatment Systems, Part A, B and C. Central Public Health and Environmental Engineering Organization, Ministry of Urban Development.

REFRENCE BOOKS:

1. Environmental Engineering Laboratory Manual by Dr. S.K. Panigrahi, L. Mohanty, S.K. Kataria& Sons

Online Learning Resources:

https://ee1-nitk.vlabs.ac.in/
https://ee2-nitk.vlabs.ac.in/